Conclusion

This paper has compared constant, deterministic, and stochastic volatility models, showing that although each can match observed option prices, they differ mathematically in how they represent uncertainty and support risk management. The comparison highlights a fundamental trade-off; where local volatility excels at matching current market prices, while stochastic volatility provides a more faithful representation of future uncertainty. For risk management applications, particularly involving tail risk estimation or dynamic hedging, stochastic volatility models offer significant advantages. Nevertheless, both models remain valuable tools, and their combined use can provide complementary insights into market behaviour and portfolio risk.

A natural extension for this paper would be to study the models at a deeper mathematical and empirical level, focusing on how each model generates return distributions and how these differ when simulated under the risk-neutral and real world probability measures. This could include analysing distributional skews implied by each model, comparing tail behaviour through simulated return paths, and examining how the geometry of the implied volatility surface evolves when parameters shift over time.

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